VNR Vignana Jyothi Institute of Engineering

Mechanical Engineering | Course Code: ME301

Applied Thermodynamics & Energy Conversion Cycles - Laboratory Manual &

Syllabus Focus: ME301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: First Law of Thermodynamics & Energy Analysis

" Syllabus Scope: Thermodynamic systems, boundary work, heat interactions, First Law for closed systems,

Unit 2: Second Law, Entropy & Exergy Destruction

" Syllabus Scope: Kelvin-Planck and Clausius statements, Carnot cycle, Clausius inequality, entropy chang

ME301: Applied Thermodynamics & Energy Conversion Cycles - Advanced Topics & Ass

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Vapor Power Cycles Rankine

" Ideal Rankine cycle, superheat, reheat cycle, regenerative feed-water heating, thermal efficiency, boiler an

Unit 4: Gas Power Cycles Otto, Diesel, Dual, Brayton

" Air standard assumptions, Otto and Diesel cycles, mean effective pressure MEP, Brayton gas turbine cycle, in

Unit 5: Refrigeration & Psychrometric Principles

" Vapor Compression Refrigeration VCR, Coefficient of Performance COP, refrigerants, psychrometric chart, spec

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Valve Timing Diagram and Port Timing Diagram of Four-Stroke and Two-Stroke Internal Combustion Eng

Lab 2: Performance Evaluation and Heat Balance Sheet of Multi-Cylinder Four-Stroke Diesel Engine

Lab 3: Determination of Coefficient of Performance COP of Vapor Compression Refrigeration Test Rig

Lab 4: Calorific Value Determination of Solid and Liquid Fuels using Bomb Calorimeter

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Yunus A. Cengel, Michael A. Boles - Thermodynamics: An Engineering Approach McGraw-Hill 8th Ed

* P.K. Nag - Engineering Thermodynamics McGraw-Hill 5th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students